

## Technology Stack

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 26 June 2026                          |
| <b>Team ID</b>       | SWTID-2026-4231                       |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 2 Marks                               |

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer                                               | Technology Chosen            | Justification / Purpose                                                               |
|------|------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------|
| 1    | <b>Frontend / Client-Side</b><br><br><i>(e.g., React, Angular, HTML/CSS)</i> | <i>HTML, CSS, JavaScript</i> | <i>Provides a lightweight, responsive interface that loads quickly for end-users.</i> |

|   |                                                                                      |                           |                                                                                        |
|---|--------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------|
| 2 | <b>Backend / Server-Side</b><br><br><i>(e.g., Node.js, Python/Django, Java)</i>      | <i>Python (Flask)</i>     | <i>Excellent integration with Python machine learning libraries like scikit-learn.</i> |
| 3 | <b>Database / Data Storage</b><br><br><i>(e.g., MySQL, MongoDB, PostgreSQL)</i>      | <i>Pandas (CSV-based)</i> | <i>Efficient for handling structured, tabular socio-economic datasets.</i>             |
| 4 | <b>Cloud / Hosting / Deployment</b><br><br><i>(e.g., AWS, Azure, Heroku, Docker)</i> | <i>Render</i>             | <i>Chosen for seamless, free-tier deployment and auto-scaling capabilities.</i>        |
| 5 | <b>Version Control &amp; CI/CD</b>                                                   | <i>GitHub</i>             | <i>Essential for team collaboration, version tracking, and code reviews.</i>           |

|          |                                                                                         |                              |                                                                            |
|----------|-----------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------|
|          | <i>(e.g., GitHub, GitLab, Jenkins)</i>                                                  |                              |                                                                            |
| <b>6</b> | <b>Third-Party APIs / Other Tools</b><br><br><i>(e.g., Stripe, Twilio, Google Maps)</i> | <i>Scikit-Learn, XGBoost</i> | <i>Used for preprocessing data and training the core regression model.</i> |